Alberta Export Growth and Diversification Potential

Question 1: Decision Brief | Priority export markets, products and diversification opportunities

Methodology

- Built two complementary evidence layers: observed 2025 Alberta domestic exports from Trade Data Online, and a forward-looking opportunity model trained on 2024 CEPII BACI HS4 trade among seven benchmark exporters. This separates “where Alberta sells now” from “where comparable trade patterns suggest room to grow.”
- Converted bilateral trade into an exporter–destination–product panel and added economic mass and trade-friction signals from CEPII Gravity (GDP, population, distance, border, language and FTA). Log transforms reduced the leverage of extreme trade values; incomplete feature rows were excluded consistently before estimation.
- Anchored the model to Alberta’s economy using its 2024 top-25 HS4 export profile, 2024 GDP at market prices ($367.7B) and provincial population. Alberta specialization acts as a capability constraint, preventing the algorithm from recommending products with no demonstrated provincial export base.
- Benchmarked Random Forest against XGBoost on the same stratified 80/20 holdout. XGBoost was retained because it reduced error and raised explained variation. Predictions are reported as unitless opportunity scores for ranking—not dollar forecasts—because Alberta itself was not an exporter in the international training panel.

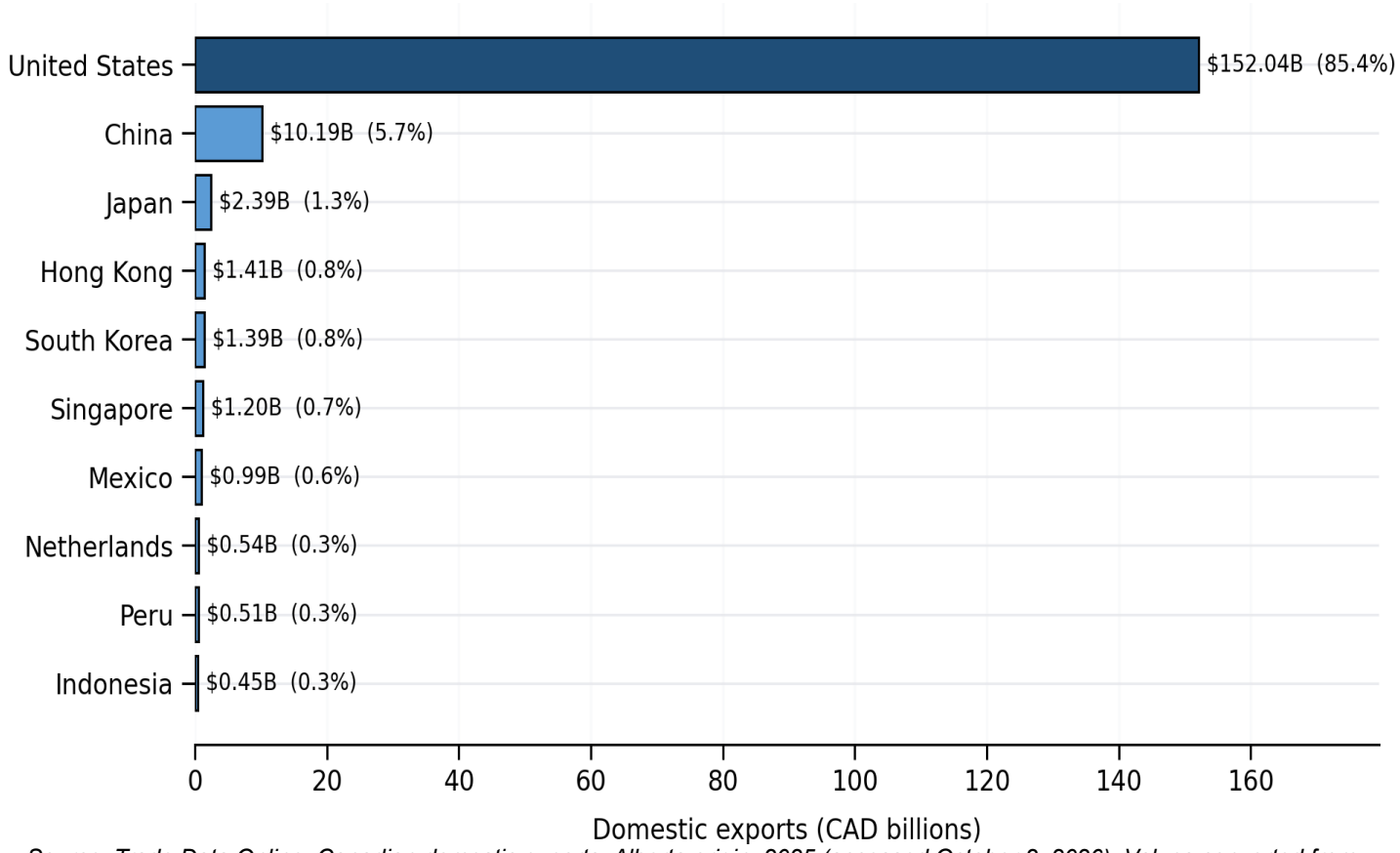
Analysis

- Alberta exported $177.96B in 2025; the United States absorbed $152.04B (85.4%). This scale is a major advantage but also the central concentration risk. China was a distant second at $10.19B; all other individual markets were below $2.4B (Attachment 1).
- The product mix confirms two diversification pathways. Energy remains the anchor, while canola, wheat, meat, polymers, pulp, sulphur and machinery create market-specific growth options. The complete destination-by-product evidence is in Attachment 2.
- XGBoost outperformed Random Forest on held-out data (R^2 0.627 vs. 0.588; RMSE 2.219 vs. 2.396 on log trade). Alberta specialization, importer GDP and product identity jointly account for 57.7% of model gain: proven capability and market scale matter more than institutional indicators alone (Attachment 3).
- The top modeled opportunities are the United States, China, Mexico, Japan, United Kingdom, South Korea, Netherlands, India, Singapore and Peru. The United Kingdom and India are the clearest diversification additions because neither was in Alberta’s observed 2025 top ten, while Hong Kong falls to 11th in modeled potential (Attachment 4).
- Recommended action: protect U.S. access, deepen high-value energy and petrochemical channels in China/Mexico/Japan, and commission buyer-level validation in the United Kingdom and India. Use the score as a screening tool; investment decisions should follow tariff, logistics, regulatory and customer-demand due diligence.

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Attachment 1 — Observed Market Concentration

Alberta's Top 10 Export Destinations, 2025



Source: Trade Data Online, Canadian domestic exports, Alberta origin, 2025 (accessed October 2, 2026). Values converted from thousands of Canadian dollars to billions. Total all countries: $177.96B.

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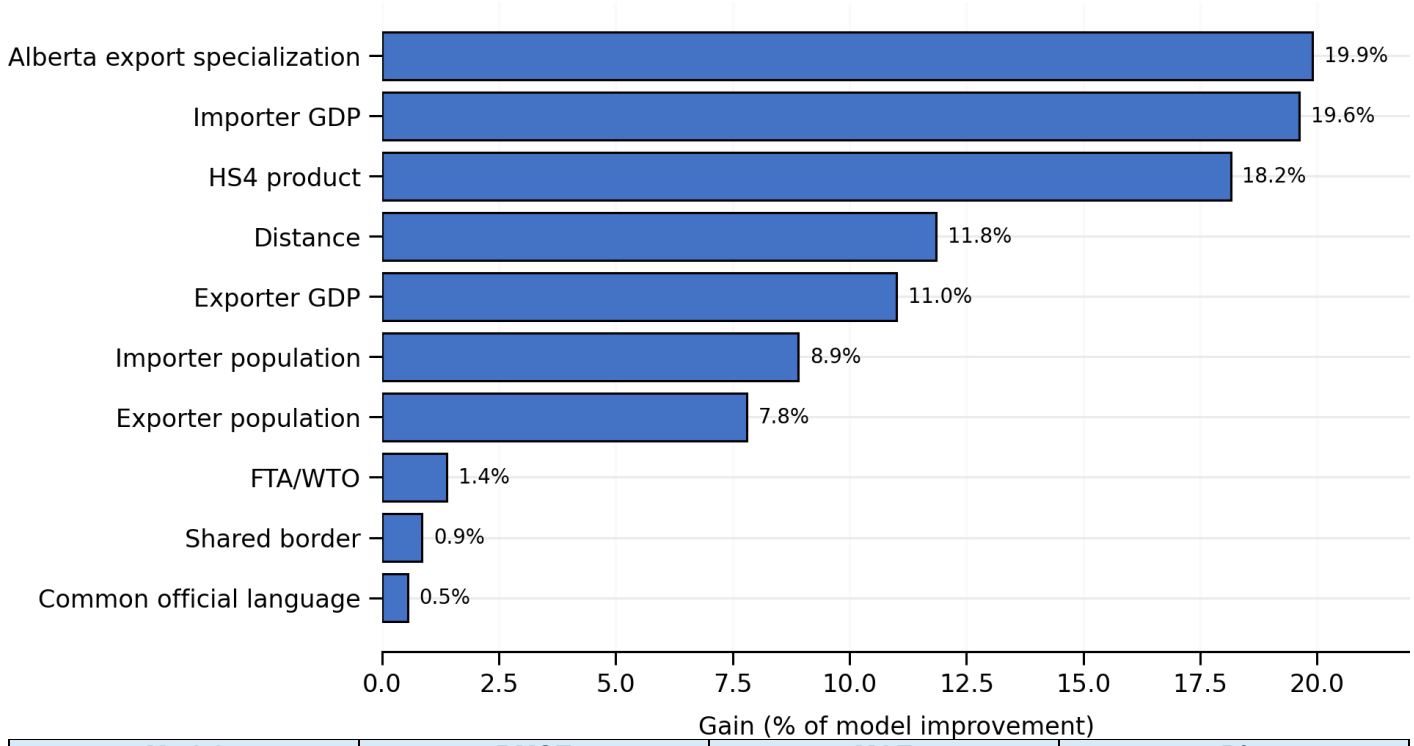
Attachment 2 — Top Five HS4 Products in Each Observed 2025 Market

Destination	HS4	Product	Value (CAD)	Destination	HS4	Product	Value (CAD)
United States	2709	Petroleum oils, crude	$110,821,617	Singapore	2709	Petroleum oils, crude	$1,088,166,780
	2711	Petroleum gases	$11,573,399		3901	Ethylene polymers	$42,054,007
	0201	Fresh/chilled bovine meat	$2,845,291		8431	Parts for machinery HS8425–8430	$17,182,329
	3901	Ethylene polymers	$2,633,250		7502	Unwrought nickel	$12,803,972
	2710	Refined petroleum oils	$2,129,225		8412	Engines and motors, n.e.s.	$4,986,717
China	2709	Petroleum oils, crude	$6,319,507	Mexico	0201	Fresh/chilled bovine meat	$231,853
	2905	Acyclic alcohols and derivatives	$611,200		1205	Canola seed	$206,314
	3901	Ethylene polymers	$588,630		1001	Wheat and meslin	$79,099
	4703	Chemical wood pulp	$437,253		0206	Edible offal	$67,108
	1205	Canola seed	$431,047		0203	Pork	$62,532
Japan	1205	Canola seed	$364,111	Netherlands	7502	Unwrought nickel	$237,632,734
	0203	Pork	$362,000		2309	Animal feed preparations	$117,699,809
	2701	Coal	$345,360		1205	Canola seed	$54,012,966
	1001	Wheat and meslin	$230,737		8105	Cobalt products	$39,899,692
	0206	Edible offal	$134,732		0201	Fresh/chilled bovine meat	$14,837,969
Hong Kong	2709	Petroleum oils, crude	$1,215,445,736	Peru	2710	Refined petroleum oils	$286,256,811
	3901	Ethylene polymers	$32,948,764		1001	Wheat and meslin	$172,856,620
	8517	Telecommunications equipment	$28,888,635		1514	Canola/rapeseed oil	$17,654,394
	7602	Aluminum waste and scrap	$25,766,303		8705	Special-purpose motor vehicles	$5,233,384
	2701	Coal	$22,700,000		8431	Parts for machinery HS8425–8430	$4,547,983
South Korea	2709	Petroleum oils, crude	$399,927	Indonesia	1001	Wheat and meslin	$258,215,425
	2701	Coal	$179,580		2503	Sulphur	$98,721,609
	7502	Unwrought nickel	$125,242		2301	Meat/fish meal	$38,511,659
	0203	Pork	$123,516		3901	Ethylene polymers	$13,344,168
	0202	Frozen bovine meat	$91,377		4705	Semi-chemical wood pulp	$7,801,859

Source: Trade Data Online, Alberta origin, 2025. Product values are Canadian dollars (not thousands). Each panel continues the same table.

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What Drives the XGBoost Opportunity Model?



Model	RMSE	MAE	R^2
Random Forest	2.396	1.904	0.588
XGBoost	2.219	1.709	0.627

Validation metrics are computed on the held-out test set and apply to log(1 + trade value). Lower RMSE/MAE and higher R^2 are better. "Gain" measures each feature's share of improvement across XGBoost tree splits.

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Attachment 4 — Modeled Export Opportunity Ranking

Rank	Destination	Opportunity score	Top five modeled HS4 opportunities, in rank order
1	United States	317.1	2710 Refined petroleum oils; 2711 Petroleum gases; 3901 Ethylene polymers; 8411 Gas turbines; 2905 Acyclic alcohols
2	China	271.5	3901 Ethylene polymers; 2710 Refined petroleum oils; 2711 Petroleum gases; 2709 Crude petroleum; 8411 Gas turbines
3	Mexico	263.0	2710 Refined petroleum oils; 8411 Gas turbines; 2711 Petroleum gases; 3901 Ethylene polymers; 2905 Acyclic alcohols
4	Japan	252.7	8411 Gas turbines; 2710 Refined petroleum oils; 2711 Petroleum gases; 3901 Ethylene polymers; 2905 Acyclic alcohols
5	United Kingdom	251.6	2710 Refined petroleum oils; 8411 Gas turbines; 2711 Petroleum gases; 2709 Crude petroleum; 3901 Ethylene polymers
6	South Korea	247.2	2710 Refined petroleum oils; 2711 Petroleum gases; 8411 Gas turbines; 3901 Ethylene polymers; 2905 Acyclic alcohols
7	Netherlands	240.1	2710 Refined petroleum oils; 2711 Petroleum gases; 8411 Gas turbines; 3901 Ethylene polymers; 2709 Crude petroleum
8	India	225.5	2710 Refined petroleum oils; 2711 Petroleum gases; 8411 Gas turbines; 2709 Crude petroleum; 3901 Ethylene polymers
9	Singapore	213.3	8411 Gas turbines; 2710 Refined petroleum oils; 2709 Crude petroleum; 2711 Petroleum gases; 3901 Ethylene polymers
10	Peru	197.8	2710 Refined petroleum oils; 3901 Ethylene polymers; 2711 Petroleum gases; 8411 Gas turbines; 2709 Crude petroleum

Note: the opportunity score is the sum of XGBoost log-scale product scores across Alberta's 2024 top-25 HS4 products. It is unitless and supports relative prioritization only; it is not a predicted dollar value. Hong Kong ranked 11th (173.5).

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